

Ghost-Free Mid-Air Imaging with a Truncated Cylindrical Array Plate

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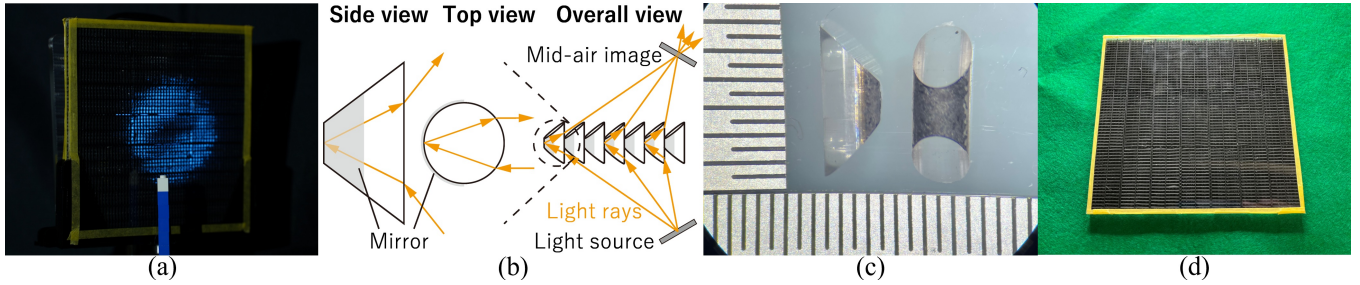


Figure 1: (a) Mid-air image formed by the TCAP. (b) Principle. (c) Enlarged photo of a truncated cylinder. (d) fabricated TCAP.

Abstract

Mid-air imaging, which forms real images in free space, is a promising augmented reality (AR) technology. However, conventional optical systems often suffer from ghost images that hinder clear observation. To address this issue, we previously proposed the Truncated Cylindrical Array Plate (TCAP), an optical structure that suppresses ghost images by combining refraction and reflection. While its effectiveness was confirmed in simulations, experimental validation remained limited. In this study, we fabricated a TCAP using acrylic cylindrical elements and demonstrated ghost-free mid-air projections of alphanumeric characters from multiple viewing angles, confirming the feasibility of the approach.

CCS Concepts

• Human-centered computing → Displays and imagers; • Hardware → Displays and imagers.

Keywords

mid-air image, optical imaging, retrotransmissive optics

ACM Reference Format:

Junpei Sano and Naoya Koizumi. 2025. Ghost-Free Mid-Air Imaging with a Truncated Cylindrical Array Plate. In *SIGGRAPH Asia 2025 Emerging Technologies (SA Emerging Technologies '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757373.3763772>

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SA Emerging Technologies '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2133-5/25/12

<https://doi.org/10.1145/3757373.3763772>

1 Introduction

To seamlessly integrate computer graphics into the real world, entirely new optical systems for visual representation are crucial. Most current AR displays, such as head-mounted devices, require users to wear specialized equipment. This often leads to discomfort due to their weight and the inconvenience associated with frequent wear and removal. To address these challenges and create a new paradigm for AR, it is necessary to develop optical elements that can be seamlessly integrated into the environment.

As one of the technologies for integrating digital content into the real world by focusing light in mid-air, Truncated Cylindrical Array Plates (TCAP) have been proposed [Sano and Koizumi 2025]. In conventional mid-air imaging technologies, the presence of ghost images—unintended spurious images—significantly degraded the visibility and usability of the mid-air imaging displays. To address this issue, TCAPs adopt a novel approach of aligning cylindrical elements, which inherently suppresses the generation of ghost images. The formation of mid-air images using this cylindrical array has been confirmed through simulations and implementations with a small number of optical elements. However, these studies were limited to proof-of-principle experiments using point sources and have not achieved mid-air imaging of graphics or characters.

In this study, we fabricated a TCAP and evaluated its function in forming mid-air images, thereby demonstrating its practical feasibility.

2 Related Work

Slit mirror arrays are among the most commonly fabricated optical elements for forming mid-air images. They generate images at positions symmetric to the light source with respect to the optical elements. However, depending on the number of reflections, some rays deviate from the intended path and become ghost images, which can interfere with image visibility.

Mid-air imaging enables image observation without physical contact, making it suitable for non-contact display applications.

Its potential has been demonstrated with large-scale mid-air displays [Kikuchi et al. 2023] and projection mapping to reduce shadows and improve visibility [Hiratani et al. 2023], underscoring its promise as a future AR display technology. However, current optical elements still suffer from issues such as ghost images, and fundamental design advances are needed. Radially arranged dihedral corner reflector array systems have been explored as a solution [Yoshimizu and Iwase 2019], but viewing positions remain limited.

3 Principle

As shown in Figure 1 (b), the TCAP consists of transparent cylindrical elements with diagonally truncated ends. A mirror is applied to the narrower side of each cut surface, forming a reflective area, while the opposite side remains transparent and serves as the entrance and exit for light. When considering the projection of the incident light rays onto the plane parallel to the TCAP, the rays are retroreflected—similar to bead-type retroreflective materials—and exit along the same path as the incoming rays. In contrast, light components along the axis perpendicular to the TCAP are not reversed and instead pass straight through.

The two-dimensional arrangement of these shaped cylinders, as illustrated in Figure 1 (b) (right), enables the formation of mid-air images. Light emitted from a display pixel is refracted and reflected by the cylinders, and reconverges at a symmetric location across the array, forming a real image in air.

The reason why this method prevents the formation of ghost images lies in the number of reflections of the image-forming light within the optical element. In previous studies using slit mirror arrays for mid-air imaging, light from the source undergoes two reflections to form a mid-air image at a position symmetric to the optical element. However, light that is reflected only once during this process results in ghost images. In contrast, our method employs only a single reflection, and therefore, image-forming light does not generate any stray paths that would lead to ghost images.

The functionality of the proposed principle has been validated through physically based rendering, which demonstrated that the cylindrical shape enables wide-angle reflection of light. As a result, it has already been shown that the mid-air image can be observed from a wider range of viewing positions compared to conventional optical elements [Sano and Koizumi 2025].

4 Fabrication

The TCAP was designed using 3DCG simulation software and fabricated by arranging diagonally cut cylindrical optical elements into a 56×60 array (vertical $\times$ horizontal). Each cylinder had a diameter of 2 mm. The acrylic cylinders were diagonally cut, mirror-coated, and custom-fabricated by Optoceramics Co., Ltd. The height along the longest edge of each cut cylinder was 5.42 mm, and the cut angle was set to 45° . These design parameters were determined using rendering simulations in Blender, a 3D computer graphics software, using its ray tracing engine, Cycles. The simulations assumed a refractive index of 1.5 (matching that of acrylic) and an observation elevation angle of 60° .

The truncated cylinders were arranged and fixed using the following procedure: First, we created an acrylic jig with sawtooth



Figure 2: Mid-air image formed by the TCAP, viewed from the left and right sides.

grooves, each row accommodating 20 diagonally cut cylinders. Next, 20 cylinders were aligned within the grooves and bonded using UV-curable resin and 3D-printed support parts. A total of 168 such sets—each consisting of 20 cylinders in a row—were prepared. Then, to uniformly arrange these rows in the same orientation, we 3D-printed V-shaped support frames that could hold eight rows of 20 cylinders each (i.e., 160 cylinders per frame). A total of 21 such frames were assembled, each populated with 160 cylinders. Finally, a 1 mm thick acrylic outer frame was fabricated, and the 20-cylinder $\times$ 8-row arrays were arranged into a 3 (horizontal) $\times$ 7 (vertical) grid, resulting in a final configuration of 60×56 cylinders. The fabricated TCAP is shown in Figure 1 (d).

5 Evaluation

To verify the applicability of the fabricated optical element for displaying characters as mid-air images, we used the letter "S" as the light source. The character was designed to be 6 cm high and 4 cm wide. The angle between the light source and the TCAP was set to 15° , and the distance between the center of the character and the TCAP—which corresponds to the depth at which the image appears to float—was set to 3.5 cm.

Figure 2 shows the mid-air image of the character captured from the left and right sides. This study demonstrates the functionality of the TCAP and highlights how rendering with 3D modeling software such as Blender can serve as a practical tool for designing real-world optical elements.

Acknowledgments

This work was supported by JST FOREST Program, Grant Number JPMJFR216L.

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